Final Project Design Phase

Computer Science - Week 11

Oct 3, 2026 - Version 0.0.4

For complete instructions and project ideas, refer to the assignment in Canvas. Specifically, review the final project requirements before you continue below. If you have not reviewed the project requirements in Canvas (you might have to go to the modules view and check the current or past week's content), you will not be able to properly complete the parts below. In addition, it may be helpful to review the information on this Drafter documentation page.

0) Project Planning and Preparation

Successful programmers spend time planning before they begin coding. *After* reviewing the project requirements, answer the following questions to help guide your design and development process.

A. What concepts from class or previous assignments might help you complete this project? Which programming concepts (loops, conditionals, lists, classes, etc.) do you expect to use? What programming skills do you want to improve while completing this project?

Many classes and a few conditionals. I want to get more comfortable with website design, since that skill is what we touched on least in class.

B. Which parts of the project do you expect to be the easiest for you? The hardest? Why?

Conditionals and function definition I expect will be relatively easy, while dealing with routes and website creation will be harder.

C. Even though you have not yet designed the final project, estimate which tasks will likely require the most time and effort? Which features or components should be implemented first?

I expect a lot of effort will be spent in designing the routes, making them as accurate to football as possible.

D. What distractions might interfere with your work and how will you minimize them? What resources will you need available while working? What questions, if any, do you still have about the assignment requirements?

The main “distraction” will be assignments for other classes, since midterms and final reviews will be happening as I work on this project. I expect to use the drafter website a lot for help regarding routes and website design.

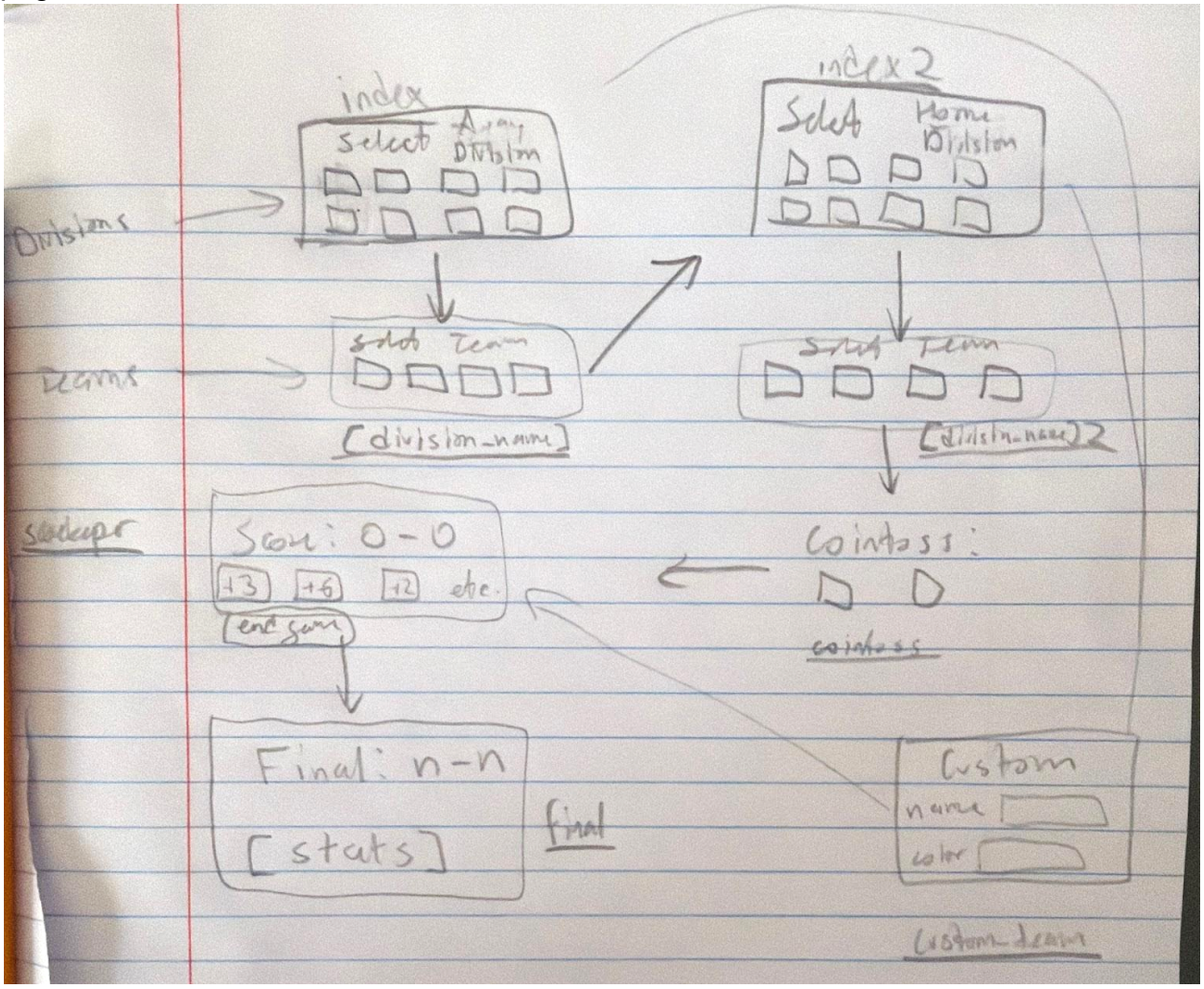
1) Site Diagram

Create a mock-up of your planned website. We suggest you make the mockup on paper, although you can also use software like PowerPoint or Slides to make a mockup. There is other design software that could also be used, but we strongly encourage you to keep it simple. Do not spend more than 15-30 minutes detailing the mock-up. Your goal should be to make a very quick and dirty diagram, to show that you have some kind of idea.

We recommend drawing each page as a rectangle, and laying out the text, buttons, and other components inside. Then, use arrows to indicate links between pages. Label each page the way you will name each function, and then put labels on the arrows to indicate what the button links. Also write down any parameters that should come into the pages.

In addition to the pages and routes, write down what fields your `State` will have, both their names and types. Think clearly about what data will need to be persistent across the entire website.

Add photo(s) or screenshot(s) of your website mock-up and labeled diagram on the next page.



2) Implementing Requirements

This table should serve as a draft for how you will implement the requirements into your final project. You may change your ideas later on, but your final site must fulfill all the requirements.

Requirement	**Implementation**
Purpose/Functionality A 1-2 sentence description of the purpose or functionality of your site:	**An easy-to-use online football score counter** Have football team selection, have buttons to update score, use team colors + names, accurately follow rules of football.
State Dataclass Attributes Minimum of 4 attributes, one of which must be a list/dict of instances of your second dataclass. Briefly describe each attribute, its type, and how it will be meaningfully modified throughout the site.	away/home team (Team), away/home scores, field goal attempts + made, touchdown conversion attempts + made, punts, turnovers (all int) The two Team instances will record the teams' names and colors, while all the int values will be used to show the current/final score and to create a final datasheet once the game ends.
Second Dataclass Your second dataclass, including its attributes, their types, and how they will be meaningfully modified (if they are modified)	**Team** Two attributes: name (str) and color (str). These are assigned to the home and away teams in state when a user either presses a button selecting a team, or writes their own using the custom feature. These data are saved in state until the user selects "new game".
Routes (minimum 7) The names of your route	index, index2, (two routes for each division and team- one home one away), away_scorekeeper, home_scorekeeper, away_touchdown, home_touchdown, …, final.

functions	
Pages (minimum 5) A brief description of the pages in your site	Index and index2 prompt the user to select a division, each division has its own page for both home and away selection, which prompts the user to select a team. The cointoss page allows the user to pick which team starts with the ball. The scorekeeper pages allow the user to input the result of each drive. The final page shows the final result with stats.
User Input (minimum 3) Description of user input components (TextBox, TextArea, CheckBox, or SelectBox) and their purposes For more information on these input types, see Drafter Documentation.	Index, index2, and each division page has buttons. The custom team pages each have two TextBox, one for name and one for color. There are also buttons on the scorekeeper pages and in the touchdown and field goal pages, asking the user if the field goal or touchdown conversion is successful or not.
If Statements (minimum 3) Brief description of what will be evaluated in your if statements and how they will meaningfully change the logic of the site	In the final page, I used if statements to make the final stat sheet be grammatically correct if certain values equal 1. I also am using if statements in a for loop to determine whether a game is considered high-scoring or not.
Loop (minimum 1) This loop must iterate over the list/dict of dataclass instances in your State attribute. Description of this loop and why it is meaningful to the site	My for loop will take in each team's score and the difference between their scores, and use that data to determine whether the game was high-scoring, and whether or not it was a lopsided result. I have not determined exactly how to do this yet. Once finished, his loop will add an extra layer of depth to the result page beyond just the score and some stats.

LLM Conversation

If you used an LLM in any part of this work, copy and paste every part of that conversation below. Otherwise, you may be given a zero for the assignment. Obviously, these words will not all be yours, and this is the exception to the requirement at the top

of this assignment that all responses be in your own words. If you did not use an LLM to assist you in this assignment, simply say so below.

No LLMs used.

Final Submission

When you are finished, download this file as a Word Document (docx) and upload the file to the appropriate assignment on Canvas.